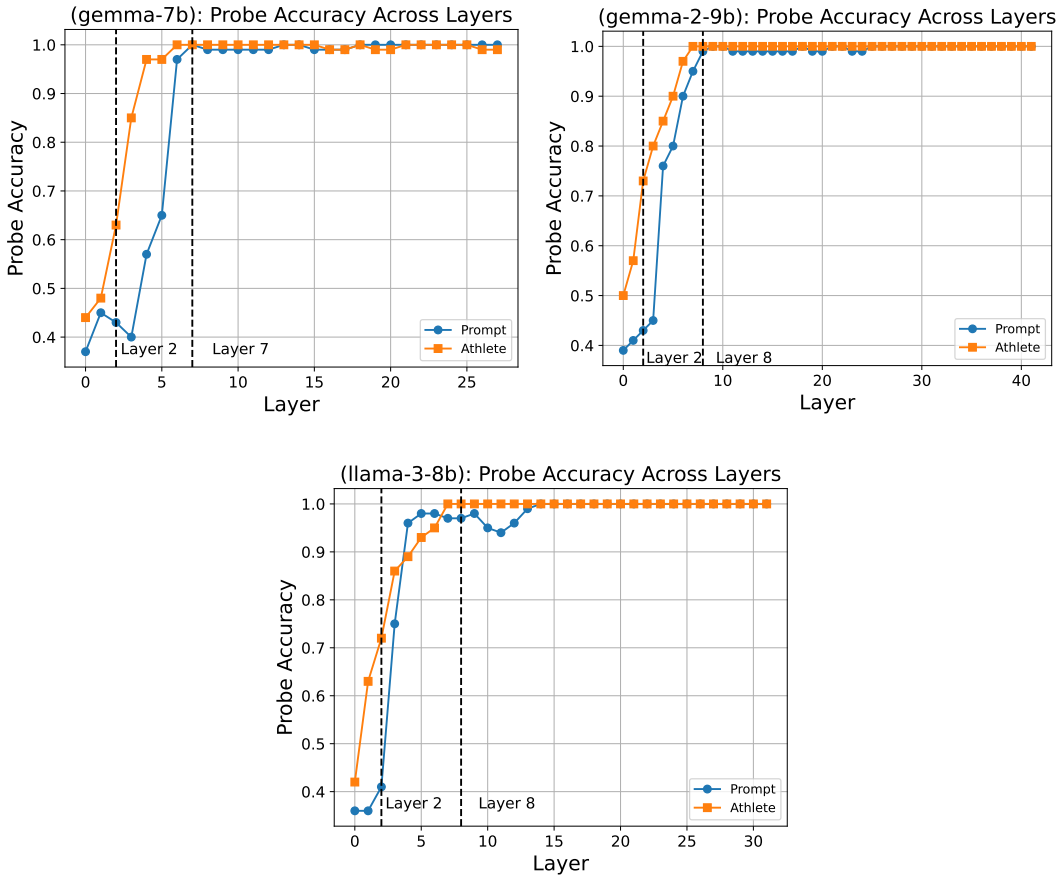

Figure 8: Accuracy of probes predicting the correct sport across layers for different models.

A.2.2 COUNTERFACT

We repeat analysis from Nanda et al. (2023) and Geva et al. (2023) on Gemma-2-9B. We first measure the effect on the difference in logits between correct and incorrect answers of facts when patching the direct path of attention heads and MLPs to the final output, shown in Figure 9. An attention head or MLP will have a large effect on the logit difference if it is important in moving the factual information to the last token position or decoding it into the correct answer. We call these components part of the "fact extraction mechanism", and aim to find the source of the factual information moved by this mechanism.

To find this source, we patch the outputs of MLPs to this "fact extraction mechanism" and measure the resultant change in logit difference (Figure 10). An MLP would cause a large change in logit difference if it caused relevant representations to form that are then moved by the "fact extraction heads" to increase the probability of the correct output. We provide the logit differences for all 64 facts along with just the first 16 facts, and see that the logit differences are similar across the dataset splits. We take the MLPs with the highest change (> 0.02) and include them in our FLU localization of CounterFact.

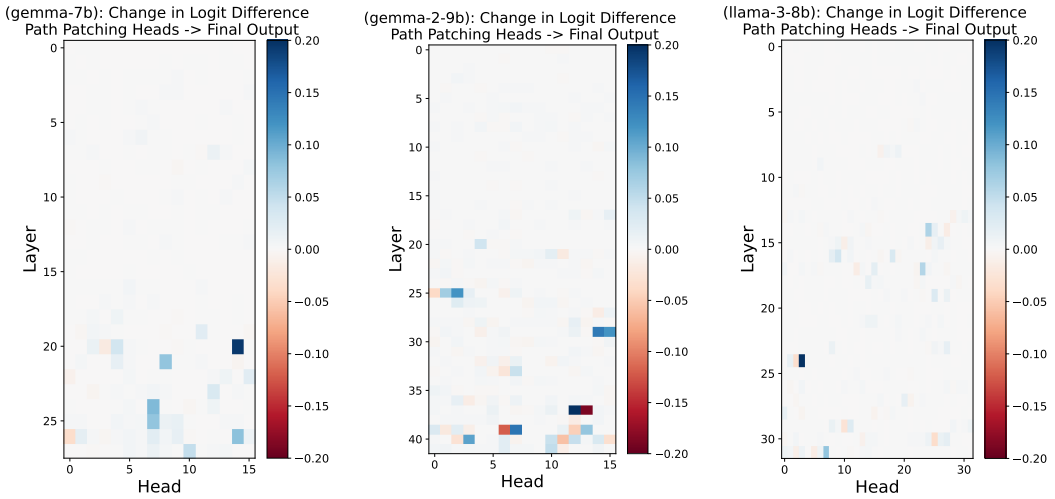

Figure 9: Change in logit difference when patching attention heads to the output.

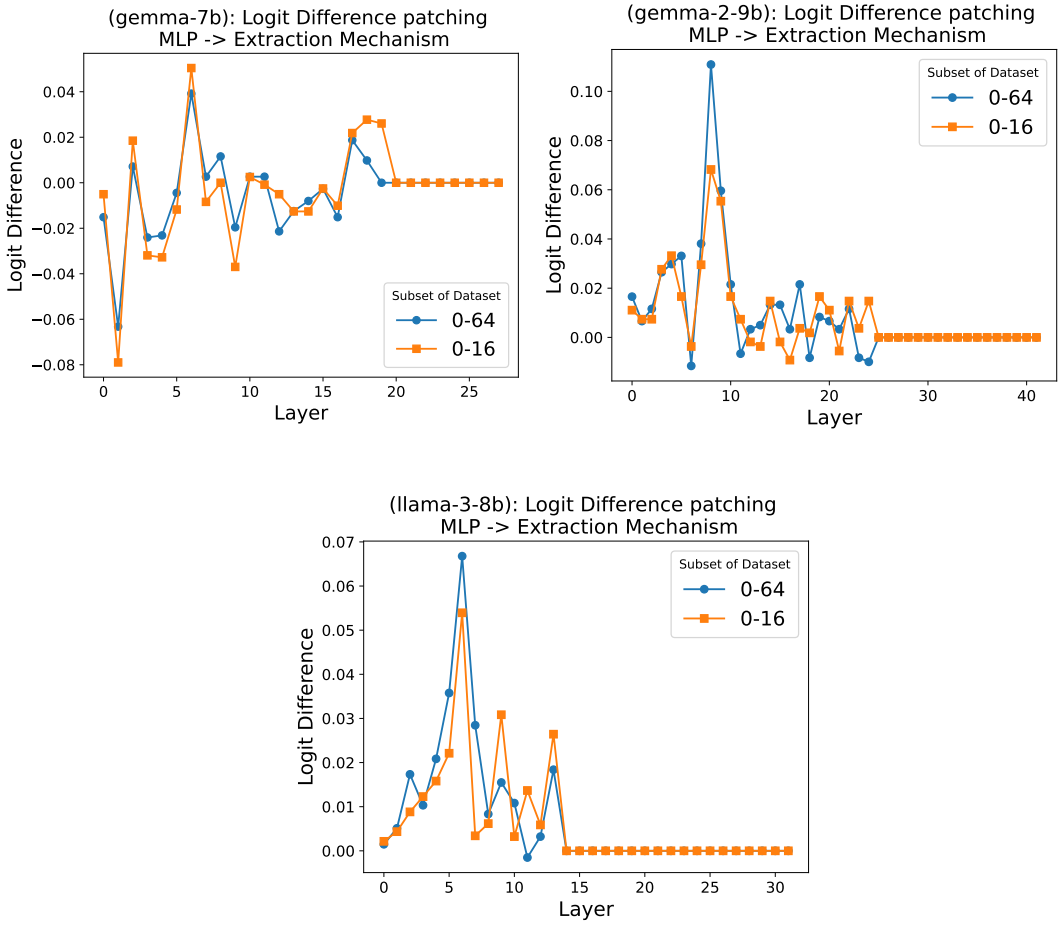

Figure 10: Logit difference when patching MLPs to the extraction mechanism found above for different models.

A.3 OT Selected Components

What MLPs do the automated OT methods localize? We explore the attribution scores of the automated localization methods (causal tracing and attribution patching) on the MLPs to see if automated localization methods can detect the FLU mechanism. In Figures 11 and 14, for Gemma-7B, we see that both CT and AP localizations target the later layer MLPs instead of the FLU mechanism (Figure 11).

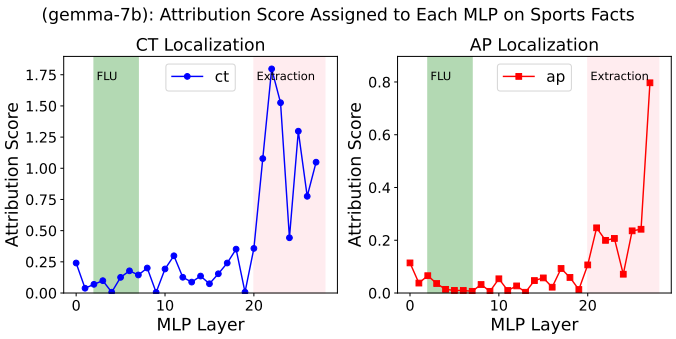

Figure 11: Attribution scores on MLPs on sports facts for Gemma-7B.

For AP localization, this trend continues with Gemma-2-9B (Figure 12) and Llama-3-8B (Figure 13). However, CT localization does highlight some of the early layer MLPs that are in the FLU mechanism, especially for Gemma-2-9B.

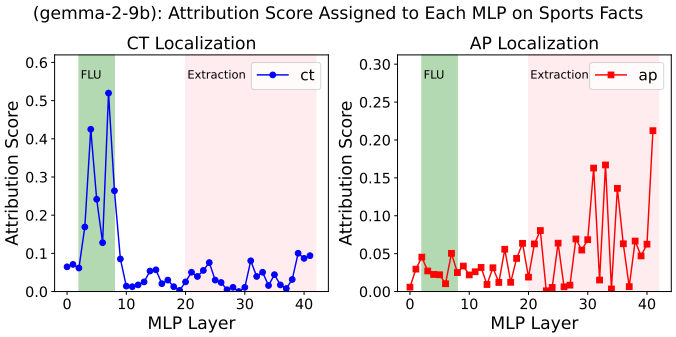

Figure 12: Attribution scores on MLPs on sports facts for Gemma-2-9B.

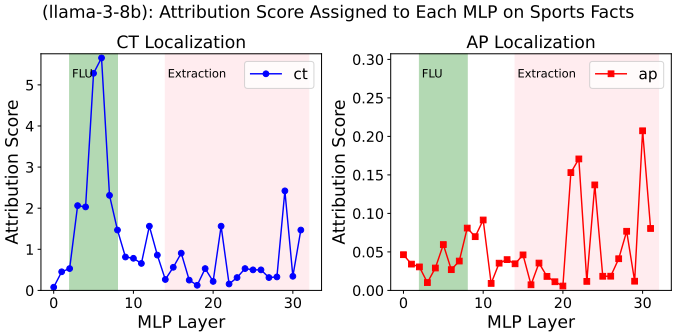

Figure 13: Attribution scores on MLPs on sports facts for Llama-3-8B.

We repeat this analysis on CounterFact in Figures 14 to 16. Again we see AP localizations in particular assign higher scores to later-layer MLPs, and CT only highlights FLU components on Gemma-2-9B, localizing other extraction layers on the other models.